Electrical Shop Management System

Introduction :

The Electrical Shop Management System is a web-based application designed to simplify the management of electrical shop products and inventory. It provides an easy-to-use platform where customers can browse available products, search for items, and view details such as price, stock availability, and product images.

The application also includes a secure administrator panel that allows authorized users to add, update, and delete products, ensuring that the inventory remains accurate and up to date. By digitizing product management, the system reduces manual effort, improves efficiency, and provides a better shopping experience for customers.

Developed using **React.js, Node.js, Express.js, and Supabase**, the application follows a modern full-stack architecture. It is responsive, user-friendly, and can be deployed on cloud platforms, making it suitable for small and medium-sized electrical stores.

This project demonstrates the practical implementation of full-stack web development concepts, including frontend design, backend API development, database management, authentication, and CRUD operations.

Objectives

- Manage electrical products efficiently.
- Allow customers to browse products without logging in.
- Provide secure login for administrators.
- Maintain product inventory.
- Display stock availability.
- Support category-wise product filtering.

Features :

Customer

- View all products
- Search products
- Filter by category
- View product details
- Check stock availability
- Responsive design

Administrator

- Secure Login
- Add Product
- Edit Product
- Delete Product
- Update Stock
- Manage Categories

Technology Stack :

Technology	Purpose
React.js	Frontend
Node.js	Backend Runtime
Express.js	REST API
Bootstrap	UI Design
Axios	API Calls
Supabase	PostgreSQL Database

Modules :

Customer Module

- Home Page
- Product Gallery
- Search Products
- Category Filter
- Pagination
- Product Details

Admin Module

- Login
- Dashboard
- Add Product
- Update Product

- Delete Product
- Logout

DATABASE DESIGN

Categories Table

Column	Type
id	Integer
name	Text

Products Table

Column	Type
id	Integer
name	Text
price	Numeric
stock	Integer
unit	Text
image	Text
category_id	Integer
is_available	Boolean

Users Table

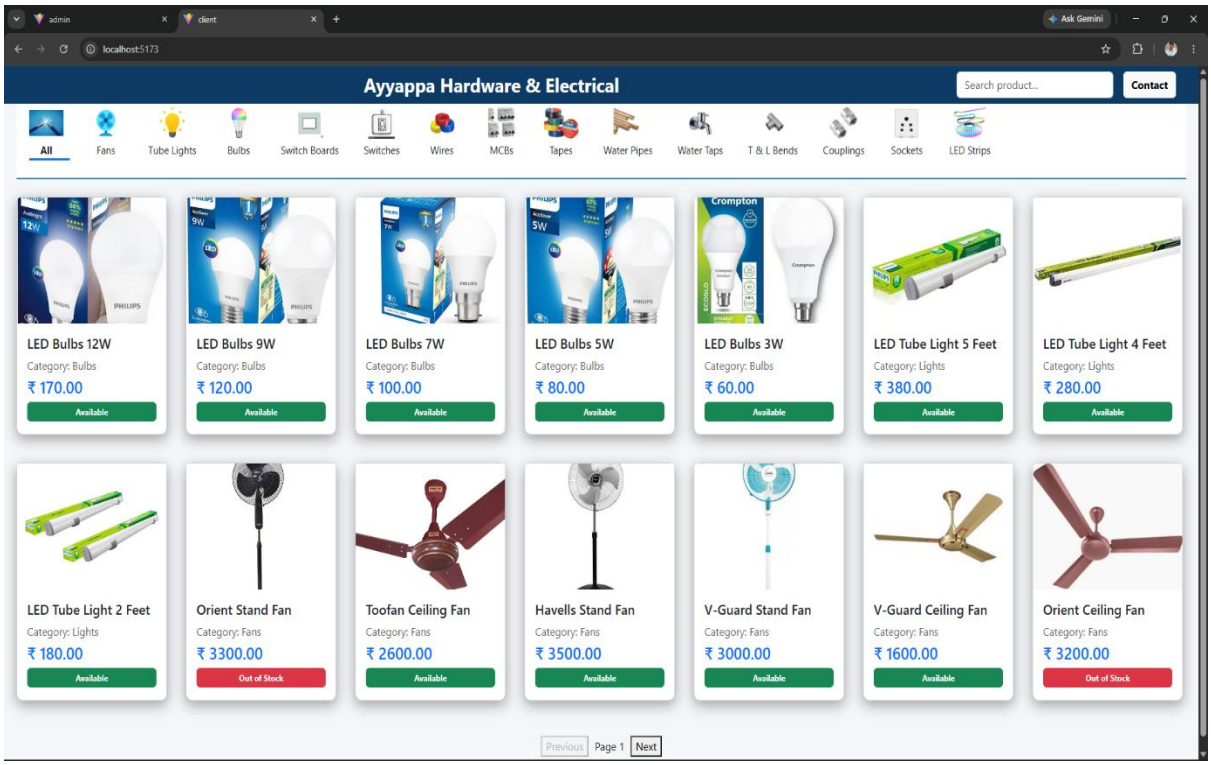
Column	Type
id	Integer
username	Text
password	Text (Encrypted)

System Workflow :

- **Customer Home Page**

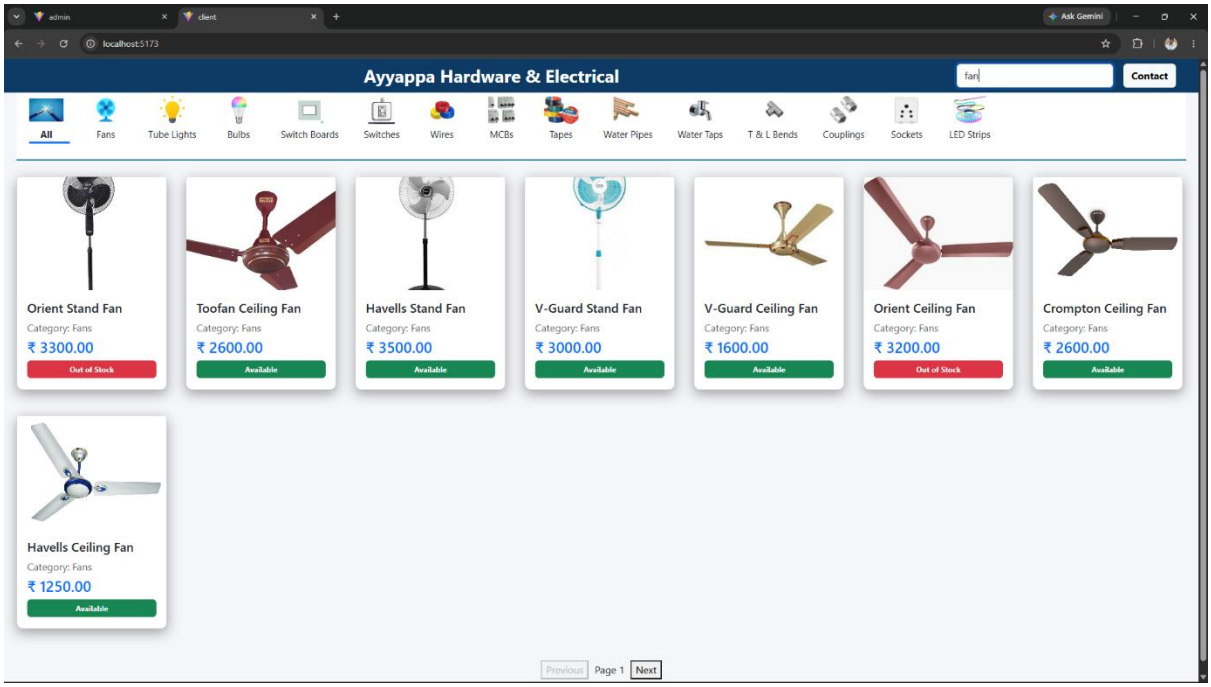
The Home Page is the first page that users see when they visit the application. It provides a clean and user-friendly interface with a navigation bar, search functionality, and product categories. Customers can easily access the available products without logging into the system.

The Browse Products page displays all available electrical products in a card-based layout. Each product card contains the product image, name, price, stock status, and unit of measurement. This layout allows customers to quickly explore the products available in the shop.



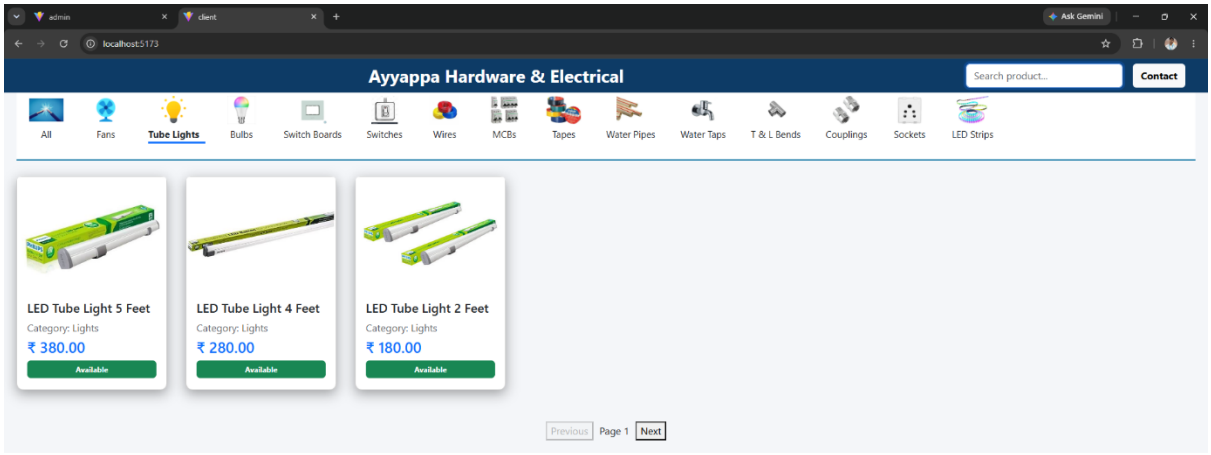
- **Search Products**

The Search feature enables customers to quickly find products by entering keywords such as the product name. As the user searches, the application displays matching products, making it easier to locate specific items.



- **Category Filter**

The Category Filter allows customers to browse products based on different categories such as Ceiling Fans, Tube Lights, Bulbs, Switches, Wires, MCBs, Water Pipes, Sockets, and other electrical items. Selecting a category displays only the products that belong to that category, improving the overall shopping experience.



- **Product Details**

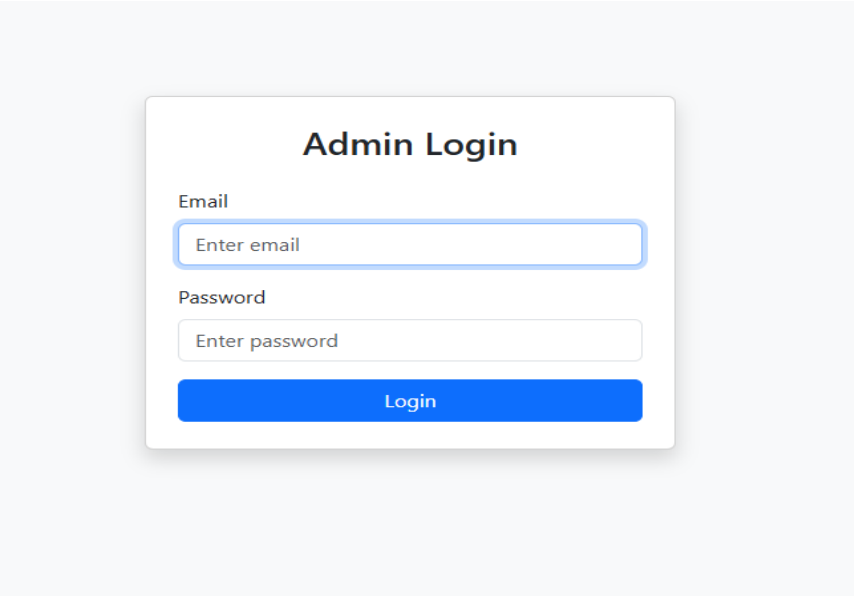
Each product card provides essential information, including the product image, product name, price, stock availability. This information helps customers understand the product before making a purchase.



Administrator Module :

- **Administrator Login**

The Administrator Login page provides secure authentication for authorized users. Only valid administrators can access the management dashboard, ensuring that product information is protected from unauthorized modifications.

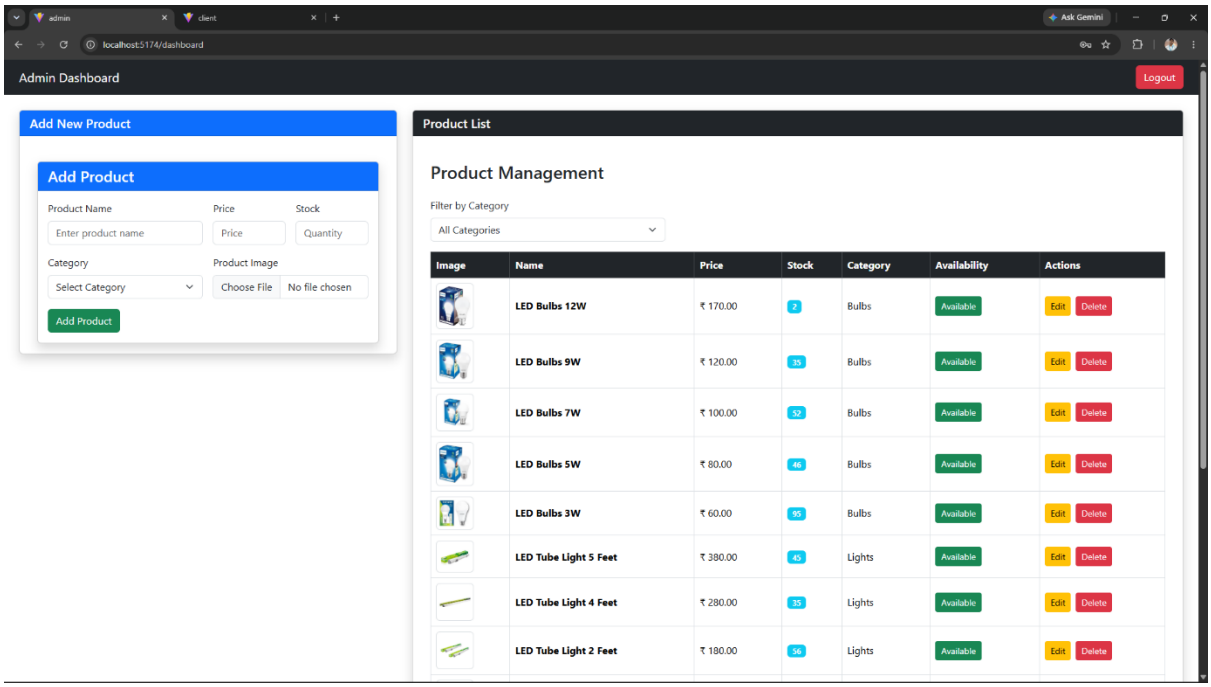


- **Administrator Dashboard**

After successful login, the administrator is redirected to the Dashboard. This page serves as the control panel for managing the shop's inventory. From here, administrators can add new products, update existing product information, and remove products from the inventory.

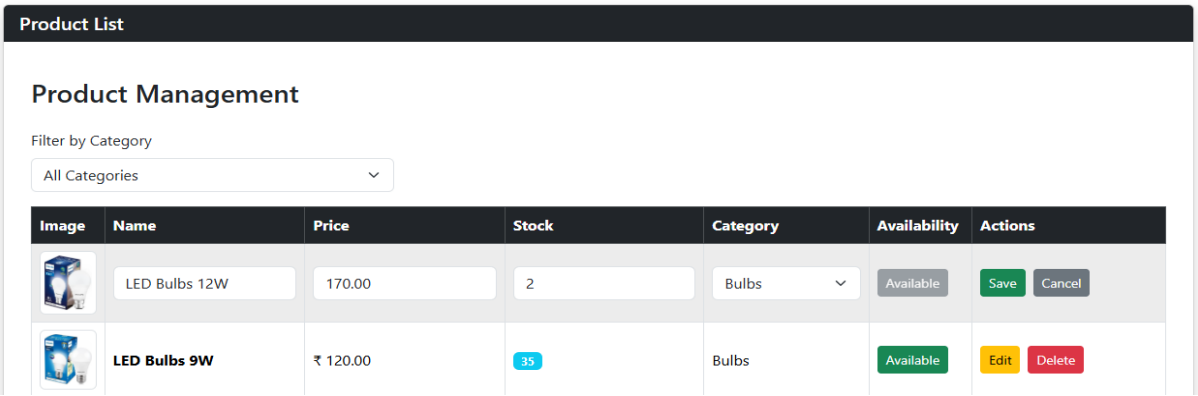
- **Add Product**

The Add Product page allows administrators to insert new products into the system. The administrator enters details such as the product name, price, stock quantity, unit, category, product image, and availability status. Once submitted, the product is stored in the database and becomes visible to customers.



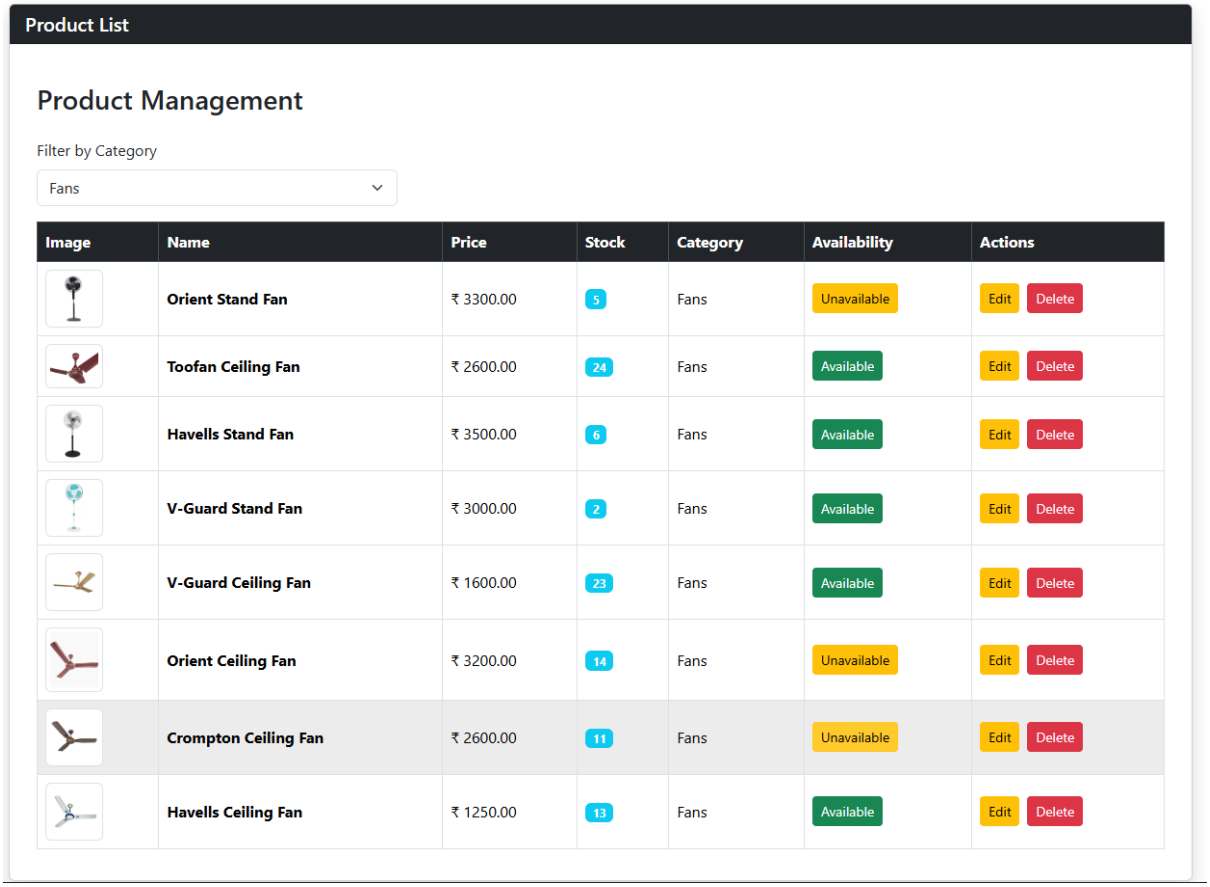
- **Update Product**

The Update Product page enables administrators to modify existing product information whenever required. Details such as price, stock quantity, category, product image, and availability can be updated to ensure that the product information remains accurate.



- **Delete Product**

The Delete Product feature allows administrators to remove products that are discontinued or no longer available. This helps maintain an accurate and up-to-date product inventory within the application.



Overall Workflow

- The Electrical Shop Management System is designed with two user roles: Customer and Administrator.
- Customers can browse products, search for specific items, and filter products by category without requiring authentication.
- Administrators, after logging in securely, can manage the inventory by performing Create, Read, Update, and Delete (CRUD) operations.
- This workflow ensures an efficient, secure, and user-friendly system for managing an electrical shop.

Conclusion :

The Electrical Shop Management System is a modern web-based application designed to streamline the daily operations of an electrical retail store. It provides customers with an easy way to browse products while enabling administrators to manage inventory securely and efficiently. Built using React, Express.js, and Supabase, the application demonstrates practical full-stack development